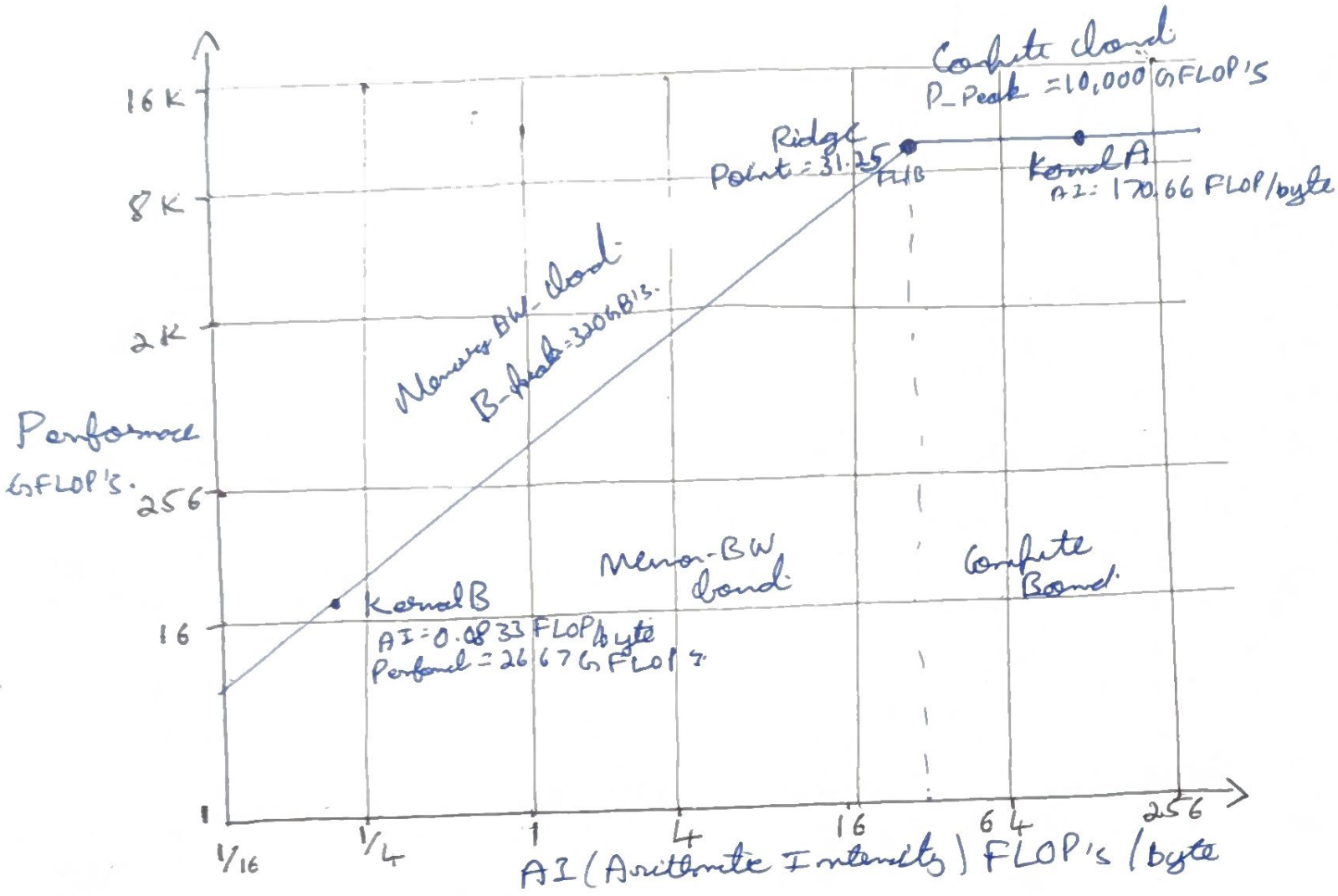

Peak Compute : 10,000 GFLOP'S

Peak Bandwidth : 320 GB'S

Ridge Point :- $\frac{\text{Peak Compute}}{\text{Peak Bandwidth}} = \frac{10,000 \text{ GFLOP'S}}{320 \text{ GBytes}} = 31.25$ FLOP/Byte

Kernal A :-

Dense GEMM

$N = 1024$

For Large N $A = N/6 = \frac{1024}{6} = 170.66$ FLOP/byte

It is Compute bound because AI (170.67 > 31.25)

Performance is capped by hardware Compute ceiling at 10,000 GFOP'S.

Kernal B :-

No. of FLOP'S = No. of additions (Length) => 4194304 FLOP'S

Bytes :- Nelements 3N×4 bytes transferred so,

$3(4194304) \times 4 = 50,331,648$ Bytes

$AI = 4194304 / 50,331648 = 0.0833$ FLOP/byte

It is memory bound because (0.0833 < 31.25)

Performance = Bandwidth × AI => 320 GB/s × 0.0833 FLOP/byte = 26.67 GFLOP'S

For Kernal A :- Because it is heavily compute bound best architectural change is is to increase no. of MACs to raise the peak TFLOPS ceiling.

For Kernal B :- Because it is memory bound effective architectural change is the increase memory bandwidth (from standard DDR mem. to High BW memory).